

on Pasqal's Orion system is important because it links quantum methods to a biologically meaningful task rather than a purely abstract benchmark.

There are also signs of experimental validation entering the picture. In research highlighted by St. Jude Children's Research Hospital in 2025, investigators reported that augmenting machine-learning-based drug discovery with quantum computing outperformed comparable classical-only models in identifying promising compounds against a KRAS target. What makes this result especially significant is the claim that the work included experimental validation, which is a stronger signal of practical relevance than simulation-only studies. At a broader industry level, analyses published in 2025 argue that life sciences could derive substantial economic value from quantum-enabled molecular modelling, particularly when combined with AI to improve in silico prediction, reduce wet-lab iterations, and generate better-quality training data for downstream models.

Still, it is important to be realistic. Quantum computing has not yet transformed the pharmaceutical industry overnight, and it is not replacing medicinal chemists, structural biologists, or classical computational platforms. Its near-term role is more targeted: improving selected calculations, enriching screening workflows, and helping researchers explore chemical hypotheses that are currently too costly or uncertain to test at scale. That incremental value, however, could be highly consequential in an industry where even modest improvements in hit identification and lead optimisation matter enormously.

Materials design: From theory to application

Materials science presents a similarly strong case for quantum computing because useful materials often depend on subtle electronic effects that are difficult to capture with

sufficient accuracy. Whether the goal is a better battery electrolyte, a more efficient catalyst, a higher-temperature superconductor, or a more durable semiconductor interface, success depends on understanding how atoms arrange themselves, how electrons move, and how those interactions change under real operating conditions. Classical modelling remains indispensable, yet many strategically important materials problems involve strongly correlated states, excited-state dynamics, or reactive surfaces that push conventional methods to their limits.

Recent case studies illustrate how quantum techniques are beginning to contribute. IBM researchers reported quantum-computer-based surface reaction calculations for lithium battery materials, applying embedding methods and hybrid quantum algorithms to model oxygen reduction reactions at electrode surfaces. The significance of this work lies in its focus on chemically relevant surface phenomena rather than isolated textbook molecules; in battery science, these interfacial reactions often determine efficiency, degradation, and safety. The researchers reported reaction energy estimates within chemical accuracy relative to high-level classical benchmarks for the selected active space, suggesting that quantum-assisted workflows can become useful for electrochemical materials research.

Another widely discussed real-world example comes from Microsoft's scientific discovery program, which in 2024 used AI and high-performance computing to screen more than 32 million candidate materials and identify a novel solid-state battery material that could be synthesized. Although this milestone was driven primarily by AI and classical computation rather than by a quantum processor, it is highly relevant to the quantum computing story because Microsoft explicitly frames chemistry and

materials discovery as a flagship future application for large-scale quantum systems. The case demonstrates the practical workflow that quantum computing is expected to enhance over time: massive candidate generation, computational narrowing, and faster validation of promising materials with better physics-based models.

Another example comes from a 2025 Canadian collaboration involving Xanadu, the University of Toronto, and the National Research Council of Canada. Their project aims to develop quantum algorithms for battery simulation that go beyond static ground-state estimation and instead address quantum dynamics more directly. That is an important shift because many useful materials problems, especially in electrochemistry, are inherently dynamical. If these algorithms mature, they could help scientists model charge transport, reaction kinetics, and degradation mechanisms with much greater fidelity than is currently practical.

These cases show that the road from theory to application is not a single dramatic leap. It is a layered transition in which classical simulation, AI, laboratory work, and quantum methods increasingly reinforce one another. In that sense, materials design may become one of the clearest demonstrations of quantum value because it sits at the intersection of fundamental physics, industrial relevance, and measurable performance outcomes.

Collaborations with industry: Pharma and energy

Industry collaboration has become one of the most important drivers in moving quantum computing for chemistry and materials science from theoretical promise towards application-oriented validation. The reason is practical: pharmaceutical and energy companies own many of the hardest molecular and materials problems, while quantum hardware providers,